Langlands duality of the affine Hecke category

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Outline

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Backgrounds

Recall the unipotent version of Local Langlands conjecture:

Conjecture

There exists a canonical finite-to-one map

$$\left\{\begin{matrix}\text{Representations of } G_k \\ \text{with a non-zero } I\text{-fixed vector}\end{matrix}\right\} \longrightarrow \left\{\begin{matrix}\text{Representations of } W_K \\ \text{factoring through } W_K \twoheadrightarrow \mathbb{Z} \to G^\vee(\mathbb{C})\end{matrix}\right\}$$

Kazhdan–Lusztig proved the following geometric interpretation of the affine Hecke algebra, hence solved the unipotent local Langlands conjecture.

Theorem (Kazhdan–Lusztig)

We have the following equivalence of algebras

$$\mathcal{H} = K^{G^\vee \times \mathbb{G}_m}(\mathrm{St}^\vee)$$

Backgrounds

Then, Bezrukavnikov proved the following categorification of the above equivalence of algebras

Theorem (Bezrukavnikov)

There is a monoidal equivalence of categories

$$\Phi_{I,I} : D(I\backslash LG/I) \to D\mathrm{Coh}^{G^\vee}\left(\widetilde{\mathcal{N}}^\vee \times^L_{\mathfrak{g}^\vee} \widetilde{\mathcal{N}}^\vee\right)$$

where $\widetilde{\mathcal{N}}^\vee \times^L_{\mathfrak{g}^\vee} \widetilde{\mathcal{N}}^\vee$ is the correct version of Steinberg here.

However, one may see that the $\mathbb{G}_m$-action is missing in this categorification.

Backgrounds

A central step in the above categorification is given by

Theorem (Arkhipov–Bezrukavnikov)

We have the following monoidal equivalence of categories

$$\Phi_{\mathcal{JW}} : D_{\mathcal{JW}}(\mathrm{Fl}) \to D\mathrm{Coh}^{G^\vee}\left(\widetilde{\mathcal{N}}^\vee\right)$$

We were able to enhance this step to add in the $\mathbb{G}_m$-action, which is the main result of the paper.

Theorem.

There is a monoidal equivalence of categories

$$\Phi_{\mathcal{JW},\,\mathrm{gr}} : D\mathrm{Coh}^{G^\vee \times \mathbb{G}_m}\left(\widetilde{\mathcal{N}}^\vee\right) \simeq D_{\mathcal{JW},\,\mathrm{gr}}(\mathrm{Fl})$$

where $D_{\mathrm{gr}}(\mathrm{Fl})$ is the category of graded sheaves introduced by Ho–Li.

Outline

The original equivalence

We first review the original equivalence of Arkhipov–Bezrukavnikov.

Construction

The construction of the functor $\Phi_{\mathcal{IW}} : D_{\mathcal{IW}}(\mathrm{Fl}) \to D\mathrm{Coh}^{G^\vee}\left(\widetilde{\mathcal{N}}^\vee\right)$ is summarized as follows:

1. Find a monoidal functor
$$\Phi_0 : \mathbf{Rep}(G^\vee \times T^\vee) \to \mathbf{Perv}_I^{\mathrm{waki}}(\mathrm{Fl})$$
2. Upgrade it to a monoidal functor
$$\Phi_1 : \mathrm{Coh}_{\mathrm{fr}}^{G^\vee \times T^\vee}\left(\widehat{\mathcal{N}}_{\mathrm{aff}}^\vee\right) \to \mathbf{Perv}_I^{\mathrm{waki}}(\mathrm{Fl})$$
3. Take derived categories
$$\Phi : D\mathrm{Coh}^{G^\vee}\left(\widetilde{\mathcal{N}}^\vee\right) \to D(\mathbf{Perv}_I(\mathrm{Fl}))$$
4. Post compose the Whittaker-averaging functor and get
$$\Phi_{\mathcal{IW}} : D\mathrm{Coh}^{G^\vee}\left(\widetilde{\mathcal{N}}^\vee\right) \to D_{\mathcal{IW}}(\mathrm{Fl})$$

The original equivalence

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1. For $\lambda \in X_*(T)$ we define the Wakimoto sheaves $J_\lambda := j_{\mu *} *_I j_{\nu !}$ for $\lambda = \mu - \nu$, with $\mu, \nu \in X_*(T)^+$. They are perverse sheaves satisfying $J_\lambda *_I J_\mu = J_{\lambda+\mu}$. Thus, we have a monoidal functor

$$\mathbf{Rep}(T^\vee) \to \mathbf{Perv}_I^{\mathrm{waki}}(\mathrm{Fl})$$

   The right category is the subcategory of $\mathbf{Perv}_I(\mathrm{Fl})$ consisting of objects admitting a filtration by Wakimoto sheaves, hence a monoidal category.

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2. Gaitsgory's central functor $Z : \mathbf{Rep}(G^\vee) \to \mathbf{Perv}_I(\mathrm{Fl})$, which is a central monoidal functor with convolution exact image. The image of Z admits a filtration by Wakimoto sheaves, so we have a monoidal functor
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They combine to give the desired monoidal functor

$$\Phi_0 : \mathbf{Rep}(G^\vee \times T^\vee) \to \mathbf{Perv}_I^{\mathrm{waki}}(\mathrm{Fl})$$

The original equivalence

The upgrade to Φ_1 relies on the following formalism:

De-equivariantization

Let H be an affine group scheme, and let $\mathcal{A}$ be a monoidal Abelian category. Then, to upgrade a monoidal functor $\mathbf{Rep}\, H \to \mathcal{A}$ to a monoidal functor $\mathbf{Mod}^H_{A,\mathrm{fr}} \to \mathcal{A}$, it suffices to give a equivariant map $A \to B$, where

$$B = \mathrm{Hom}^{\mathrm{comm},\otimes}_{\mathrm{Ind}(\mathcal{A})}(1, F(\mathcal{O}(H)))$$

Here comm and $\otimes$ denote certain constraints on the map.

Then, Φ_1 relies on the following application of the above principle on Φ_0:

The original equivalence

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Then, Φ_1 relies on the following application of the above principle on Φ_0:

1. Gaitsgory's central functor is equipped with a nilpotent tensor endomorphism

$$M : Z \to Z$$

2. There is a highest weight arrow $Z(V_\lambda) \to j_{\lambda *} = J_\lambda$ compatible with the Plücker relations, and that its composition with M is 0.

The original equivalence

Then, to upgrade to $\Phi : D\mathrm{Coh}^{G^\vee}\left(\widetilde{\mathcal{N}}^\vee\right) \to D(\mathbf{Perv}_I(\mathrm{Fl}))$, note first that since $\hat{N}^\vee_{\mathrm{aff}}/G^\vee \times T^\vee$ is smooth, we have $D\mathrm{Coh}^{G^\vee \times T^\vee}\left(\widetilde{\mathcal{N}}^\vee\right) = \mathrm{Perf}^{G^\vee \times T^\vee}\left(\widetilde{\mathcal{N}}\right)$.

The functor $K(\Phi_2) : D\mathrm{Coh}^{G^\vee \times T^\vee}\left(\widehat{\mathcal{N}}^\vee_{\mathrm{aff}}\right) \to D\left(\mathbf{Perv}^{\mathrm{waki}}_I(\mathrm{Fl})\right)$ then vanishes on the boundary $\widehat{\mathcal{N}}^\vee_{\mathrm{aff}} - \widehat{\mathcal{N}}^\vee$. Thus, it factors through the quotient category $D\mathrm{Coh}^{G^\vee \times T^\vee}\left(\widehat{\mathcal{N}}^\vee\right)$, giving Φ.

Finally, we post-compose with the Whittaker-averaging functor to get $\Phi_{\mathcal{IW}}$. Here, Whittaker-averaging is the functor $\mathrm{Av}_{\mathcal{IW}} : D(\mathbf{Perv}_I(\mathrm{Fl})) \to D_{\mathcal{IW}}(\mathrm{Fl})$ given by convolution with the Iwahori–Whittaker sheaf at the unit Δ_e.

The original equivalence

The equivalence is finally proven via the computation of stalks

$$(Z_{\mathcal{JW}}(V):\Delta_\lambda)=(Z_{\mathcal{JW}}(V):\nabla_\lambda)=\dim\left(V^{T^\vee=\lambda}\right)$$

The computation of cohomology

$$\begin{aligned}&\mathrm{Hom}_{D\mathrm{Coh}^{G^\vee}(\widetilde{\mathcal{N}}^\vee)}\left(\mathcal{O}_{\widetilde{\mathcal{N}}^\vee},V\otimes\mathcal{O}_{\widetilde{\mathcal{N}}^\vee}(\lambda)[n]\right)\\ \hookrightarrow\ &\mathrm{Hom}_{D_{\mathcal{JW}}(\mathrm{Fl})}\left(\Phi_{\mathcal{JW}}\left(\mathcal{O}_{\widetilde{\mathcal{N}}^\vee}\right),\Phi_{\mathcal{JW}}\left(V\otimes\mathcal{O}_{\widetilde{\mathcal{N}}^\vee}(\lambda)\right)[n]\right)\end{aligned}$$

and finding the two separate sets of generators

$$\{\mathcal{O}(\lambda),\lambda\in X_*(T)\},\{V\otimes\mathcal{O}(\lambda),V\in\mathbf{Rep}\,G^\vee,\lambda\in X_*(T)^+\}$$

for $D\mathrm{Coh}^{G^\vee}\left(\widetilde{\mathcal{N}}^\vee\right)$ and the generator $\{J_\lambda\}$ for $D_{\mathcal{JW}}(\mathrm{Fl})$.

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The original equivalence

The graded version

The graded version

To upgrade the above functor to an equivalence, we need a suitable sheaf theory. This is the graded sheaf theory. The main properties we will use is the following.

Theorem

For any stack X one may associate it to a sheaf theory $D_{\mathrm{gr}}(X)$ with the following properties:

1. $D_{\mathrm{gr}}(X)$ is a triangulated category with the perverse t-structure and a transversal weight structure.
2. There is an associated graded functor $\mathrm{gr} : D_{\mathrm{mixed}}(X) \to D_{\mathrm{gr}}(X)$ and a forgetful functor $\mathrm{oblv}_{\mathrm{gr}} : D_{\mathrm{gr}}(X) \to D(X)$. The functors preserves and reflects the respective perverse t-structures. gr preserves the weights.
3. For nice enough morphisms of stacks $f : X \to Y$ we can define adjunction $f^* \dashv f_*$ and $f_! \dashv f^!$ on graded sheaves. These functors are compatible with gr and $\mathrm{oblv}_{\mathrm{gr}}$.
4. We have the decomposition

$$\mathrm{Hom}\big(\mathrm{oblv}_{\mathrm{gr}}(\mathcal{F}), \mathrm{oblv}_{\mathrm{gr}}(\mathcal{G})\big) \simeq \bigoplus_{n\in\mathbb{Z}} \mathrm{Hom}_{\mathrm{gr}}(\mathcal{F}, \mathcal{G}\langle n\rangle)$$

The graded version

To upgrade $\Phi_{\mathcal{JW}}$ to the graded version, the only obstruction lies in lifting

$$\Phi_{0,\,\mathrm{gr}} : \mathbf{Rep}(G^\vee \times T^\vee \times \mathbb{G}_m) \to \mathbf{Perv}^{\mathrm{waki}}_{I,\,\mathrm{gr}}(\mathrm{Fl})$$

to

$$\Phi_{1,\,\mathrm{gr}} : \mathrm{Coh}^{G^\vee \times T^\vee \times \mathbb{G}_m}_{\mathrm{fr}}\left(\widehat{\mathcal{N}}^\vee_{\mathrm{aff}}\right) \to \mathbf{Perv}^{\mathrm{waki}}_{I,\,\mathrm{gr}}(\mathrm{Fl})$$

this is obtained by taking the associated graded of Gaitsgory's central functor in the mixed setting $M : Z \to Z(-1)$. The Plücker relations and the composite to 0 conditions are still satisfied, and the rest of the construction is analogous.

The graded version

Finally, to prove the graded equivalence, we simply apply the diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\mathbb{G}_m}(\mathcal{F},\mathcal{G}) \arrow[r] \arrow[d, hook] & \mathrm{Hom}_{\mathrm{gr}}\big(\Phi_{\mathcal{JW},\,\mathrm{gr}}(\mathcal{F}),\Phi_{\mathcal{JW},\,\mathrm{gr}}(\mathcal{G})\big) \arrow[d, hook] \\
\bigoplus_{n\in\mathbb{Z}} \mathrm{Hom}_{\mathbb{G}_m}(\mathcal{F},\mathcal{G}\langle n\rangle) \arrow[r, "\simeq"] \arrow[d, "\simeq"'] & \bigoplus_{n\in\mathbb{Z}} \mathrm{Hom}_{\mathrm{gr}}\big(\Phi_{\mathcal{JW},\,\mathrm{gr}}(\mathcal{F}),\Phi_{\mathcal{JW},\,\mathrm{gr}}(\mathcal{G})\langle n\rangle\big) \arrow[d, "\simeq"] \\
\mathrm{Hom}\big(\mathrm{oblv}_{\mathbb{G}_m}(\mathcal{F}),\mathrm{oblv}_{\mathbb{G}_m}(\mathcal{G})\big) \arrow[r, "\simeq"] & \mathrm{Hom}\big(\mathrm{oblv}_{\mathrm{gr}}(\Phi_{\mathcal{JW},\,\mathrm{gr}}(\mathcal{F})),\mathrm{oblv}_{\mathrm{gr}}(\Phi_{\mathcal{JW},\,\mathrm{gr}}(\mathcal{G}))\big)
\end{tikzcd}$$

which proves fully faithfulness. Essential surjectivity is analogous to the previous proof.

Thank you!

Questions?